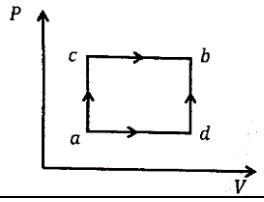
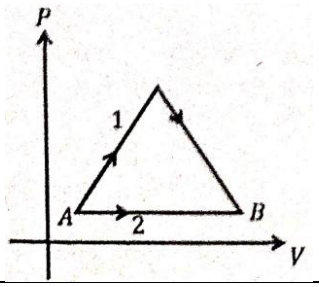
Question	The first law of thermodynamics is based on:
Type	multiple choice
Option	The concept of temperature.
Option	The concept of conservation of energy
Option	The concept of working of heat engine.
Option	The concept of entropy.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	When a system is moved from state a to state b along the path ach, it is discovered that the system absorbs 200 J of heat and performs 80 J of work. Along the path adb, heat absorbed Q = 144 J. The work done along the path adb is: 
Type	multiple choice
Option	6 J
Option	12 J
Option	18 J
Option	24 J
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A system is taken from state A to state B along two different paths, 1 and 2. If the heat absorbed and work done by the system along these two paths are Q_1, Q_2 and W_1, W_2 respectively, then:
Type	multiple choice
Option	$Q_1 = Q_2$
Option	$W_1 = W_2$
Option	$Q_1 - W_2 = Q_2 - W_2$
Option	$Q_1 + W_1 = Q_2 + W_2$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If a gas undergoes the change in its thermodynamic state from A to B via two different paths, as shown in the given pressure (P) versus volume (V) graph, then: 
Type	multiple choice

Option	The temperature of the gas decreases in path 1 from A to B.
Option	The heat absorbed by the gas in path 1 is greater than in path 2.
Option	The heat absorbed by the gas in path 2 is greater than in path 1.
Option	The change in internal energy in path 1 is greater than in path 2.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In a given process, dW = 0, dQ < 0, then for the gas:
Type	multiple choice
Option	Temperature increases
Option	Volume decreases
Option	Pressure decreases
Option	Pressure increases
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	110 joule of heat is added to a gaseous system, whose internal energy changes by 40 J. Then the amount of work done is
Type	multiple choice
Option	150 J
Option	70 J
Option	110 J
Option	40 J
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If we had a system on which 10 J of work is done and the internal energy change of the system is 3.5 J, in this case, the amount of heat loss by the system will be?
Type	multiple choice
Option	-8.5 J
Option	-6.5 J
Option	-2.85 J
Option	-13.5 J
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A system X is neither in thermal equilibrium with Y nor with Z. The system Y and Z
Type	multiple choice
Option	May be in thermal equilibrium
Option	Cannot be in thermal equilibrium
Option	Must be in thermal equilibrium
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A gas is compressed from a volume of $2m^3$ to a volume of $1m^3$ at a constant pressure of $100\ N/m^3$. Then it is heated by supplying some heat at constant

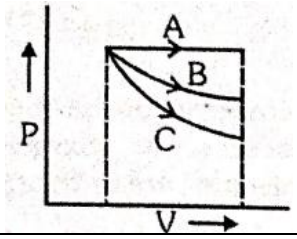
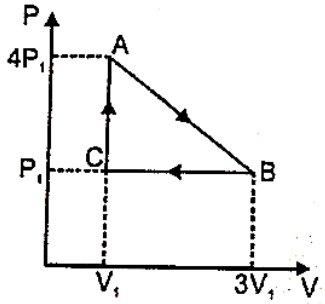
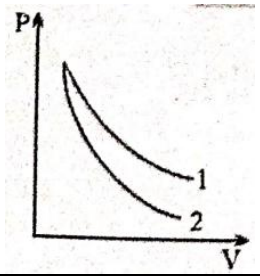
	volume. Total Heat of 150 J is given to the gas. As a result, the internal energy of the gas:
Type	multiple choice
Option	Increases by 250 J
Option	Increases by 50J
Option	Decreases by 250 J
Option	Decreases by 50 J
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Zeroth law of thermodynamics
Type	multiple choice
Option	Deals with mass and energy
Option	Deals with heat engines
Option	States that if two systems are both in equilibrium with a third system, they are in thermal equilibrium with each other
Option	Does not exist
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	An ideal gas is compressed to half its initial volume by means of several processes. Which of the following process results in the maximum work being done on the gas?
Type	multiple choice
Option	Adiabatic
Option	Isobaric
Option	Isochoric
Option	isothermal
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its temperature. The ratio of C_p / C_v for the gas is equal to:
Type	multiple choice
Option	$4/3$
Option	2
Option	$5/3$
Option	$3/2$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In thermodynamic processes, which of the following statements is not true?
Type	multiple choice
Option	In an adiabatic process the system is insulated from the surroundings
Option	In an isochoric process the pressure remains constant.
Option	In an isothermal process the temperature remains constant.
Option	In an adiabatic process, $PV^{\gamma} =$ constant.
Answer	2
Solution	

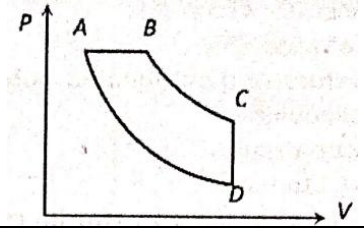
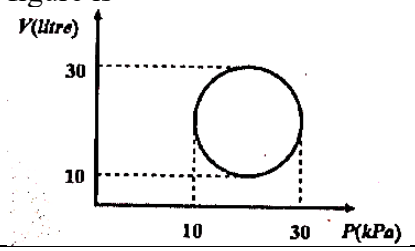
Positive Marks	4
Negative Marks	1
Question	If the ratio of specific heat of a gas at constant pressure to that at constant volume is γ, the change in internal energy of a mass of gas, when the volume changes from V to 2V constant pressure p, is
Type	multiple choice
Option	$R/(\gamma-1)$
Option	pV
Option	$pV/(\gamma-1)$
Option	$\gamma pV/(\gamma-1)$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A monoatomic gas of n- moles is heated from temperature T_1 to T_2 under two different conditions (i) at constant volume (ii) at constant pressure. The change in internal of the gas is
Type	multiple choice
Option	More for (i)
Option	More for (ii)
Option	Same in both cases
Option	Independent of number of moles
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	For an isothermal expansion of a perfect gas, the value of $\frac{\Delta P}{P}$ is equal -
Type	multiple choice
Option	$-\gamma^{1/2}\frac{\Delta V}{V}$
Option	$-\frac{\Delta V}{V}$
Option	$-\gamma\frac{\Delta V}{V}$
Option	$-\gamma^2\frac{\Delta V}{V}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	An ideal gas at 27°C is compressed adiabatically to $\frac{8}{27}$ of its original volume. If $\gamma=\frac{5}{3}$, then the rise in temperature is
Type	multiple choice
Option	450 K
Option	375 K
Option	225 K

Option	405 K
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	During the adiabatic expansion of 2 mole of a gas, the internal energy of the gas is found to decrease by 2 joules, the work done during the process on the gas will be equal to -
Type	multiple choice
Option	1 J
Option	$-1J$
Option	$2J$
Option	$-2J$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A gas expands under constant pressure 10. 5. 6. P from volume V_1 to V_2. The work done by the gas is
Type	multiple choice
Option	$P(V_2 - V_1)$
Option	$P(V_1 - V_2)$
Option	$P(V_1^\gamma - V_2^\gamma)$
Option	$P\frac{V_1V_2}{V_2 - V_1}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is incorrect?
Type	multiple choice
Option	In an isobaric process, $\Delta P = 0$
Option	In an isobaric process, $\Delta W = 0$
Option	In an isobaric process, $\Delta T = 0$
Option	In an isobaric process, $\Delta Q = 0$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	According to the second law of thermodynamics:
Type	multiple choice
Option	heat energy cannot be completely converted to work
Option	work cannot be completely converted to heat energy
Option	for all cyclic processes we have dQ/T<0
Option	the reason all heat engine efficiencies are less than 100% is friction, which is unavoidable
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

Question	A Carnot engine working between 300 K and 600 K has work output of 800 J per cycle. What is amount of heat energy supplied to the engine from source per cycle
Type	multiple choice
Option	1800 J/cycle
Option	1000 J/cycle
Option	2000 J/cycle
Option	1600 J/cycle
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Efficiency of Carnot engine is 100% if
Type	multiple choice
Option	$T_{\sin k} = 273\,K$
Option	$T_{\sin k} = 0\,K$
Option	$T_{source} = 273\,K$
Option	$T_{source} = 0\,K$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A Carnot's engine used first an ideal monoatomic gas then an ideal diatomic gas. If the source and sink temperature are 411°C and 69°C respectively and the engine extracts 1000 J of heat in each cycle, then area enclosed by the PV diagram is
Type	multiple choice
Option	100 J
Option	300 J
Option	500 J
Option	700 J
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The temperature of reservoir of Carnot's engine operating with an efficiency of 70% is 1000 K. The temperature of its sink is
Type	multiple choice
Option	300 K
Option	400 K
Option	500 K
Option	700 K
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A carnot engine shows efficiency of 40% on taking energy at 500 K. To increase the efficiency to 50% at what temperature it should take energy?
Type	multiple choice
Option	400 K
Option	700 K
Option	600 K

Option	800 K
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A Carnot's engine is made to work between 200°C and 0°C first and then between 0°C and -200°C. The ratio of efficiencies of the engine in the two cases is
Type	multiple choice
Option	$1.73:1$
Option	$1:1.73$
Option	$1:1$
Option	$1:2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	An ideal refrigerator has a freezer at a temperature of -13°C. The coefficient of performance of the engine is 5. The temperature of the air (to which heat is rejected) will be
Type	multiple choice
Option	$325^0 C$
Option	$325 K$
Option	$39^0 C$
Option	$320^0 C$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In a mechanical refrigerator, the low temperature coils are at a temperature of $-23^0 C$ and the compressed gas in the condenser has a temperature of 27°C. The theoretical coefficient of performance is
Type	multiple choice
Option	5
Option	8
Option	6
Option	6.5
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The first operation involved in a Carnot cycle is
Type	multiple choice
Option	Isothermal expansion
Option	Adiabatic expansion
Option	Isothermal compression
Option	Adiabatic compression
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	During isothermal, isobaric and adiabatic processes, work done for same change

	in volume will be maximum for: 
Type	multiple choice
Option	Isothermal
Option	Isobaric
Option	Adiabatic
Option	None of the above
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	An ideal gas is taken round the cycle ABCA. In the cycle the amount of work done involved is: 
Type	multiple choice
Option	$12\,P_1V_1$
Option	$6\,P_1V_1$
Option	$3\,P_1V_1$
Option	P_1V_1
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	P-V plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond respectively to 
Type	multiple choice
Option	He and O_2
Option	O_2 and He
Option	He and Ar
Option	O_2 and N_2
Answer	2
Solution	

Positive Marks	4
Negative Marks	1
Question	In pressure - volume diagram given below, the isochoric, isothermal, and isobaric parts respectively, are 
Type	multiple choice
Option	BA, AD, DC
Option	BA, CB, DC
Option	AB, BC, CD
Option	CD, DA, AB
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Heat energy absorbed by a system in going through a cycle process shown in figure is 
Type	multiple choice
Option	$10\pi J$
Option	$5\pi J$
Option	$10^2\pi J$
Option	$10^3\pi J$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If the mean free path of atoms is doubled then the pressure of the gas will become:(P is initial pressure)
Type	multiple choice
Option	$P/4$
Option	$P/2$
Option	$P/8$
Option	P
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A rigid diatomic has how many degrees of freedom
Type	multiple choice
Option	3
Option	4
Option	5
Option	6

Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	V_{rms}, V_{av}, V_{mp} are root mean square, average and most probable speeds of molecules of a gas obeying Maxwellian velocity distribution. Which of the following statements is correct?
Type	multiple choice
Option	$V_{rms} < V_{av} < V_{mp}$
Option	$V_{rms} > V_{av} > V_{mp}$
Option	$V_{rms} < V_{av} < V_{mp}$
Option	$V_{rms} > V_{av} < V_{mp}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	For a gas, the r.m.s speed at 800 K is:
Type	multiple choice
Option	Four times the value at 200 K
Option	Half the value at 200 K
Option	Twice the value at 200 K
Option	Same as at 200 K
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	For a gas, $\frac{R}{C_V} = 0.67,$ the gas is made up of molecules which are:
Type	multiple choice
Option	Diatomic
Option	Mixture of diatomic and polyatomic molecules
Option	Monoatomic
Option	Polyatomic
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	One liter of gas A and two liters of gas B, both having the same temperature 100°C and the same pressure 2.5 bar will have the ratio of average kinetic energies of their molecules as:
Type	multiple choice
Option	$1:1$
Option	$1:2$
Option	$1:4$
Option	$4:1$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	n_1 mole of monoatomic gas is mixed with n_2 mole of diatomic gas such that

	$\gamma_{mix} = 1.5,$ then:
Type	multiple choice
Option	$n_1 = 2n_2$
Option	$n_2 = 2n_1$
Option	$n_1 = n_2$
Option	$n_1 = 3n_2$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A gas mixture consist of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is:
Type	multiple choice
Option	4 RT
Option	15 RT
Option	9 RT
Option	11 RT
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The respective speeds of the molecules are 1, 2, 3, 4 and 5km/sec. The ratio of their r.m.s velocity and the average velocity will be
Type	multiple choice
Option	$\sqrt{11}:3$
Option	$3:\sqrt{11}$
Option	$1:2$
Option	$3:4$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	At a given temperature the root mean square velocities of oxygen and hydrogen molecules are in the ratio
Type	multiple choice
Option	$16:1$
Option	$1:16$
Option	$4:1$
Option	$1:4$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1